Key ceremony — operator sheet (one process, run three times)

The reasoning and review history are in `CEREMONY.md`. The steps are done by `tessera-ceremony.sh`, which you read before running. Every command you type is on its own line: type one, press Enter, read the result, then the next.

Run	L	YubiKey	Encrypted backup stick	Public stick
1 rehearsal	REH	C	volume name REH-PRIV	volume name REH-PUB
2 root	ROOT	A	ROOT-PRIV	ROOT-PUB
3 release	REL	B	REL-PRIV	REL-PUB

Stick labels are yours (what each holds). The short names are only the **volume names** typed into *Disks*. The script reaches you on the **INPUT** stick: a new stick that has **never** been in the NUC, written by the Pangolin in a separate, networked Tails session (below). The NUC is not in the path.

If a step prints FAIL, nothing has been deleted: fix what it says and run the same step again. Never reboot between `generate` and `tests` unless the owner says so: the key is in memory until it is on the card and backed up. Never photograph or type into chat a PIN or passphrase. **[PHOTO]** = photograph the screen. **[WRITE]** = write on the public sheet.

Before the session: write the INPUT stick (a separate Tails boot, network ON)

On the Pangolin, boot Tails **with** networking (it connects through Tor). No YubiKey, no other stick inserted. When Tor is connected, open a terminal:

```
git clone https://github.com/fsgeek/tessera.git
```

```
cd tessera
```

```
git checkout <the commit the owner names>
```

```
sha256sum docs/ceremony/tessera-ceremony.sh
```

It must equal the owner's hash. **[WRITE]** it, in groups of 8. *(If `git` is not found: in Tor Browser open `https://raw.githubusercontent.com/fsgeek/tessera/<commit>/docs/ceremony/tessera-ceremony.sh`, save it to the Tor Browser folder, and run `sha256sum` on the saved file instead; then copy that file in the step below.)* Insert the new INPUT stick; in *Disks*, format it (FAT, volume name `INPUT`); open it in *Files*; then:

```
cp docs/ceremony/tessera-ceremony.sh /media/amnesia/INPUT/
```

```
sha256sum /media/amnesia/INPUT/tessera-ceremony.sh
```

Same hash. Eject; shut down. This session is amnesic: nothing from it persists except the INPUT stick.

Part 1 (one boot)

Welcome Screen: **set an administration password**; do **not** unlock Persistent Storage. Network off. Insert the INPUT stick, open it in *Files*, open a terminal.

0. Check and read the script

```
sha256sum /media/amnesia/INPUT/tessera-ceremony.sh
```

It must equal the paper. Then read it (space pages, q quits):

```
less /media/amnesia/INPUT/tessera-ceremony.sh
```

Copy it into memory:

```
cp /media/amnesia/INPUT/tessera-ceremony.sh /tmp/c.sh
```

`sha256sum /tmp/c.sh`

Same hash again. Eject INPUT and remove it. From here on, every step is `bash /tmp/c.sh <step>`.

1. Start [PHOTO] [WRITE] GnuPG version

`bash /tmp/c.sh start ROOT`

(REH, ROOT or REL.)

2. Card — insert the run's YubiKey only. **[WRITE]** serial.

`bash /tmp/c.sh card`

3. PINs — the script shows what to type.

`bash /tmp/c.sh pins`

4. P-256 slot

`bash /tmp/c.sh slot`

5. Generate — enter the passphrase twice when asked. **[WRITE]** the fingerprint, then read it back.

`bash /tmp/c.sh generate`

6. Backup — first, in *Disks*: the `L-PRIV` stick → ⋮ *Format Disk* (GPT) → + → *Internal disk for use with Linux systems only (Ext4)*, volume name `ROOT-PRIV` (etc.), *Password protect volume (LUKS)*, the key's passphrase. Open it in *Files*. Then: **[WRITE]** LUKS type and cipher.

`bash /tmp/c.sh backup`

It asks for the passphrase once more (the restore test). On PASS, eject `L-PRIV` and remove it.

7. Public stick — in *Disks*: the `L-PUB` stick → *Format Disk* (GPT) → + → *For use with all systems (FAT)*, volume name `ROOT-PUB` (etc.), no password. Open it in *Files*. Then: **[WRITE]** the hash.

`bash /tmp/c.sh public`

Leave it in.

8. Onto the card — the script shows what to type.

`bash /tmp/c.sh tocard`

9. Policy [PHOTO] (the `S UIF-1` line)

`bash /tmp/c.sh policy`

10. Tests [PHOTO] — the first signature needs a touch; the second must not be touched. **[WRITE]** the error text.

`bash /tmp/c.sh tests`

11. Check the public stick yourself (typed by hand, not by the script). **[PHOTO]**

`ls -la /media/amnesia/ROOT-PUB`

Want: `key.pub.asc`, `proof.txt`, `proof.txt.asc`, nothing else.

`gpg --show-keys /media/amnesia/ROOT-PUB/key.pub.asc`

Want: a line starting `pub` and **no line starting `sec`**, and your fingerprint. Then eject, remove, shut down.

Part 2 (a new boot)

Welcome Screen as before. INPUT stick: repeat step 0 (check, read, copy to `/tmp/c.sh`). Open `L-PRIV` in *Files* (passphrase). Have the YubiKey ready but not inserted.

`bash /tmp/c.sh verify ROOT`

Type the fingerprint from the sheet; insert the YubiKey when asked; PIN and touch; remove it when asked; then the passphrase for the backup. Want PASS. Eject; shut down.

After each run

- **Run 1:** reset C (`gpg --card-edit`, then `admin`, `factory-reset`, `y`, `yes`, `q`).
- **Runs 2–3:** the YubiKey and its `L-PRIV` stick go to **different places**; the passphrase paper to a third. A's and B's things are not stored together. Tell the owner only "done". The `L-PUB` stick goes to the owner and never returns to a signing session.
- **From step 8 on, never reset the card**, whatever fails. Keep the card and the backup and bring the photos.